Examiner only

SECTION B

Answer **all** *questions in the spaces provided.*

7. Sian was given the following instructions for an experiment to find the enthalpy change for the reaction below.

$$Mg(s) + 2HCl(aq) \longrightarrow MgCl_2(aq) + H_2(g)$$

Instructions

- Weigh the magnesium.
- Place 25.0 cm^3 of 0.5 mol dm^{-3} hydrochloric acid in an insulating plastic container.
- Measure the initial temperature of the acid in the container.
- Add the magnesium to the acid and stir.
- Measure the highest temperature reached.

(a) Draw a labelled diagram showing how the apparatus is set up to carry out this experiment. You should show how the heat losses could be minimised. [3]

Examiner only

(b) Sian recorded the following data.

Mass of magnesium strip = 0.1 g

Initial temperature of acid = 21.0 °C

Highest temperature reached = 35.5 °C

Use these data to calculate the enthalpy change for the reaction, Δ*H*.

$$Mg(s) + 2HCl(aq) \longrightarrow MgCl_2(aq) + H_2(g)$$

Assume that 4.18 J is needed to raise the temperature of 1.0 cm^3 of all aqueous solutions by 1.0 °C and that the hydrochloric acid was in excess.

Give your answer in kJ mol^{-1}. [3]

2410U201
07

Δ*H* = ... kJ mol^{-1}

(c) Efan repeated the experiment using 0.20 g of magnesium and 50.0 cm^3 of 0.1 mol dm^{-3} hydrochloric acid. Although he used his data correctly to calculate the enthalpy change of reaction for 1 mol of magnesium, his answer was numerically much less than that obtained by Sian. By using the data above and that in part *(b)*, explain why Efan obtained a different value for Δ*H*. [2]

....

....

Examiner only

(d) Sian used a one-decimal place balance to weigh the magnesium. Calculate the maximum percentage error in the mass of magnesium she used. Show clearly how you obtained your answer. [2]

Maximum percentage error = ... %

(e) Burning magnesium, when placed in a gas-jar containing chlorine, forms magnesium chloride and energy is released.

$$Mg(s) + Cl_2(g) \longrightarrow MgCl_2(s)$$

Suggest why this reaction cannot be used to measure the enthalpy change for the reaction that forms magnesium chloride from magnesium. [1]

....

....

....

11

Examiner only

SECTION B

Answer **all** *questions in the spaces provided.*

7. *(a)* Ethanol, C_2H_5OH, is a liquid at room temperature. Write the equation that corresponds to the standard enthalpy change of formation of ethanol. [2]

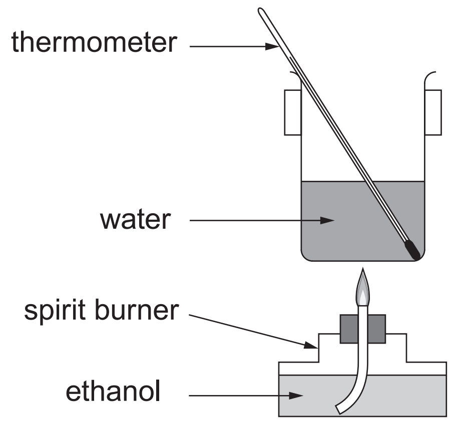
....

(b) Why is it difficult to measure the standard enthalpy change of formation of ethanol directly? [1]

....

....

(c) The following apparatus can be used to determine the enthalpy change of combustion of ethanol, $\Delta_c H$.



(i) Explain how the enthalpy change of combustion is calculated from the results collected using this apparatus. [3]

....

....

....

....

....

....

2410U201
05

Examiner only

(ii) Why is it difficult to use this apparatus to determine the enthalpy change of combustion of ethane, C_2H_6? [1]

....

....

(d) Two students carried out an experiment using the apparatus in part *(c)*.

Neither followed the given method correctly. Euan extinguished the flame after only 10 seconds. Carys continued to heat for two minutes after the water had boiled.

Explain the effect that each of the students' errors had on the values they obtained in this experiment. [2]

Effect of Euan's error

....

....

Effect of Carys' error

....

....

Examiner only

(e) Standard enthalpy changes of formation, $\Delta_f H^\theta$, can be found by using standard enthalpy changes of combustion, $\Delta_c H^\theta$.

The table shows some standard enthalpy changes of combustion.

Substance	Standard enthalpy change of combustion / kJ mol^{-1}
carbon, C	–394
hydrogen, H_2	–286
ethanol, C_2H_5OH	–1371

(i) Use these data to calculate the standard enthalpy change of formation of ethanol. Show clearly how you carried out your calculation. [3]

$\Delta_f H^\theta$ = ... kJ mol^{-1}

(ii) Amir said that hydrogen is the best fuel of these three in terms of the energy released by burning equal masses. Is he correct? Justify your answer. [2]

....

....

....

2410U201
07

14

Examiner only

SECTION B

Answer **all** *questions in the spaces provided.*

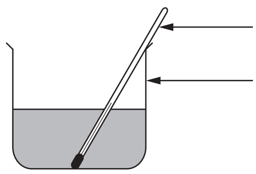
7. *(a)* A student was asked to find the enthalpy change of reaction, ΔH, for the thermal decomposition of calcium hydroxide.

$$Ca(OH)_2(s) \longrightarrow CaO(s) + H_2O(g)$$

This enthalpy cannot be measured directly so Hess's Law is generally used to calculate its value from the enthalpy changes for the reactions of calcium oxide and calcium hydroxide with hydrochloric acid.

The student added a known mass of calcium oxide to hydrochloric acid and measured the temperature change. This was repeated using a known mass of calcium hydroxide. In each experiment $50.0\,cm^3$ of $1.40\,mol\,dm^{-3}$ hydrochloric acid was used.

The diagram shows the apparatus used.



When 1.90 g of calcium oxide was used a temperature rise of 20.5 °C was observed.

(i) The equation for the reaction of calcium oxide with hydrochloric acid is shown.

$$CaO(s) + 2HCl(aq) \longrightarrow CaCl_2(aq) + H_2O(l)$$

Use this equation to show that the hydrochloric acid was in excess in the reaction with calcium oxide. [3]

....

....

Examiner only

(ii) Calculate the energy released in this reaction between calcium oxide and hydrochloric acid. [1]

Energy released = ... J

(iii) Calculate the molar enthalpy change for the reaction between calcium oxide and hydrochloric acid. [2]

$$CaO(s) + 2HCl(aq) \longrightarrow CaCl_2(aq) + H_2O(l)$$

Enthalpy change = ... kJ mol^{-1}

sign value

Examiner only

(iv) The value of the molar enthalpy change of reaction for the reaction between calcium hydroxide and hydrochloric acid is –196 kJ mol^{-1}.

Use Hess's Law and your answer to part (iii) to calculate the enthalpy change of reaction for the decomposition reaction.

$$Ca(OH)_2(s) \longrightarrow CaO(s) + H_2O(g)$$

Show clearly how you obtained your answer.

(If you do not have an answer in part (iii), assume that the enthalpy change of reaction is –110 kJ mol^{-1}. This is **not** the correct value.) [2]

Enthalpy change = ... kJ mol^{-1}

(v) Suggest **two** changes to the apparatus shown that would give a more accurate value for the enthalpy changes of the reactions. Give a reason for your answer in both cases. [2]

....

....

....

....

Examiner only

(b) Methanol, CH_3OH, is a liquid at room temperature and its molar enthalpy of combustion, $\Delta_c H$, can be found directly.

(i) Write the equation corresponding to the standard enthalpy change of combustion of methanol. [1]

....

(ii) Draw and label suitable apparatus as it is being used in an experiment to determine the enthalpy change of combustion of methanol. [2]

2410U201
07

Examiner only

(c) The equation for the combustion of ethene is shown.

$$C_2H_4(g) + 3O_2(g) \longrightarrow 2CO_2(g) + 2H_2O(g) \qquad \Delta_cH^\theta = -1387\ \text{kJ mol}^{-1}$$

(i) Use this and the values of the average bond enthalpies in the table to calculate the average bond enthalpy of C—H. [4]

Bond	Average bond enthalpy / kJ mol^{-1}
C═C	614
O═O	495
C═O	799
O—H	465

Average bond enthalpy = ... kJ mol^{-1}

(ii) State why the apparatus you have drawn in part *(b)* cannot be used to determine the enthalpy change of combustion of ethene. [1]

....

....

18

Examiner only

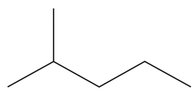
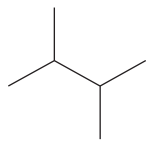
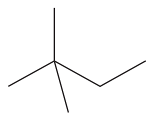
7. Petroleum ether (50–70) is a mixture of different alkanes extracted from crude oil which is commonly used as an organic solvent. The major components of petroleum ether (50–70) are the structural isomers of C_6H_{14}.

(a) (i) Give the meaning of the term ‘structural isomer’. [1]

....

....

(ii) Complete the table below showing important information about the isomers of C_6H_{14}. [3]

Name	Shortened structural formula	Skeletal formula	Boiling temperature / °C
hexane	$CH_3CH_2CH_2CH_2CH_2CH_3$		69
2-methylpentane			62
3-methylpentane	$CH_3CH_2CH(CH_3)CH_2CH_3$		63
	$(CH_3)_2CHCH(CH_3)_2$		58
2,2-dimethylbutane	$CH_3C(CH_3)_2CH_2CH_3$		50

Examiner only

(iii) State the relationship between the boiling temperature and the carbon chain length. Explain this relationship in terms of intermolecular forces. [2]

(b) Hexane can be used as a fuel in a combustion reaction.

(i) Write an equation for the complete combustion of hexane in excess oxygen. [2]

(ii) The enthalpy change of combustion ($\Delta_c H^\theta$) for hexane is approximately $-4160\,kJ\,mol^{-1}$. Explain why the enthalpy change of combustion for the isomers of hexane should be similar. [2]

(iii) 2,2-dimethylbutane is the isomer of C_6H_{14} which ignites most readily. Suggest a reason for this. [1]

2410U201 09

Examiner only

(iv) When hexane burns in a limited supply of oxygen it undergoes a different reaction known as incomplete combustion:

$$C_6H_{14} \quad + \quad 6.5O_2 \longrightarrow 6CO \quad + \quad 7H_2O$$

The bond enthalpy values for the bonds present in these molecules are given below:

Bond	Average bond enthalpy / $kJ\,mol^{-1}$
C — C	348
C — H	413
O ═ O	495
C ≡ O (in CO)	1072
O — H	464

I. Using a Hess cycle or otherwise, calculate the enthalpy change of this reaction. [3]

enthalpy change = ... $kJ\,mol^{-1}$

II. **Use the enthalpy values** from parts (b)(ii) and (b)(iv) I. to explain quantitatively why it is important to maintain an excess of oxygen while burning hexane as a fuel. [2]

....

....

....

Examiner only

III. State a health hazard associated with the incomplete combustion of hexane. [1]

....

....

17

2410U201 11